

Provia
By
Project Team



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1. Introduction

Project-Based Learning (PBL) is an educational methodology that allows students to gain practical experience through real-world projects. Managing student groups, proposals, supervisors, evaluations, and project records manually can become difficult and inefficient.

Traditional paper-based methods often result in delayed communication, misplaced records, and lack of proper tracking. To solve these issues, PROVIA was developed as a centralized web-based system for managing the complete PBL workflow.

The system helps institutions automate project management processes and improve coordination between students, supervisors, evaluators, and administrators.

1.1 Background

Many institutions still rely on manual systems for handling project proposals, evaluations, supervisor assignments, and student communication. These methods consume time and increase the possibility of errors.

As the number of students and projects increases, institutions require a digital solution capable of managing:

- Student groups
- Proposal submissions
- Supervisor assignments
- Evaluation processes
- Communication channels
- File management
- Project tracking

PROVIA was developed to provide an efficient, organized, and scalable solution for these requirements.

2. Objectives of Study

The objectives of this project are:

2. To develop a centralized PBL project management system.
3. To implement secure role-based authentication.
4. To allow online proposal submission and management.
5. To facilitate supervisor assignment and monitoring.
6. To provide evaluation and feedback mechanisms.
7. To improve communication between stakeholders.

8. To maintain project history and workflow tracking.
9. To reduce manual paperwork and administrative effort.
10. To provide a responsive and user-friendly interface.
11. To ensure secure storage of project information.

3. Methodology

The development of PROVIA followed a structured Software Development Life Cycle (SDLC) consisting of planning, design, development, and testing phases.

3.1 Planning Phase

During this phase, system requirements were collected and analyzed. The major stakeholders included:

- Students
- Supervisors
- Evaluators
- PBL Managers
- Chairpersons

The following system requirements were identified:

- Secure login system
- Group management
- Proposal submission
- Supervisor allocation
- Evaluation tracking
- Communication system
- Dashboard analytics
- File upload and attachment support

Technology selection was also performed during this phase. PHP and MySQL were selected because they are reliable, scalable, and widely used for web application development.

3.2 Design Phase

The system architecture was designed using a modular approach. Separate modules were created for students, managers, evaluators, supervisors, and administrators.

The user interface design focused on:

- Responsive layouts

- Simple navigation
- Dashboard-based interaction
- Clean user experience

The database schema was designed using relational database principles to maintain data integrity and efficient storage.

3.3 Development Phase

The system was implemented using:

- PHP for backend development
- MySQL for database management
- HTML and CSS for structure and styling
- Bootstrap for responsive design
- JavaScript for interactivity

Backend Development

The backend implementation included:

- Authentication and session management
- Proposal management
- Role-based access control
- Evaluation processing
- Database operations
- File management

Frontend Development

The frontend was designed to provide a modern and responsive user experience through dashboards, navigation menus, forms, and interactive components.

Database Development

The database contains tables for:

- Users
- Departments
- Programs
- Semesters
- Groups
- Proposals
- Evaluations

- Messages
- Notifications
- Attachments

Relationships between tables were implemented using primary and foreign keys.

3.4 Testing Phase

Several testing methods were used to ensure system reliability.

Unit Testing

Each module was tested independently.

Integration Testing

The interaction between modules was verified.

Interface Testing

The responsiveness and usability of the system were checked.

Database Testing

Database queries and relationships were tested for consistency.

Security Testing

Authentication and access permissions were tested to prevent unauthorized access.

4. Results and Findings

The developed system successfully implemented the planned functionalities and improved project management efficiency.

4.1 Role-Based Access Control

The system supports multiple user roles including:

- Student
- Supervisor
- Evaluator
- PBL Manager
- Chairman

Each role has separate permissions and dashboards, improving security and organization.

Student Features

- Create groups
- Submit project proposals
- Upload files
- Track proposal status
- View supervisor feedback
- Access communication features

Supervisor Features

- Review proposals
- Provide feedback
- Monitor assigned groups
- Submit evaluations

Manager Features

- Manage users
- Assign supervisors
- Monitor project workflows
- View analytics

Evaluator Features

- Evaluate projects
- Submit grades and recommendations

4.2 Query Submission and Routing

The system supports structured workflows for proposal submission and approval. Student submissions are routed to supervisors and managers for review.

This process ensures:

- Organized workflow management
- Faster communication
- Proper approval tracking
- Reduced paperwork

4.3 Status Tracking and History

The application maintains project status records and history logs. Users can track proposal progress and monitor updates throughout the project lifecycle.

Benefits include:

- Better transparency
- Improved monitoring
- Easier record management
- Accountability of actions

4.4 Feedback and Attachments

The system allows supervisors and evaluators to provide feedback directly through the platform. Students can upload project-related documents and attachments securely.

Supported features include:

- File uploads
- Comments and feedback
- Document management
- Evaluation attachments

4.5 Database Schema

The relational database structure efficiently stores and manages all project-related information.

Key database advantages include:

- Data consistency
- Efficient retrieval
- Secure storage
- Relationship management

5. Discussion

5.1 Effectiveness of the Hierarchical Workflow

The hierarchical workflow implemented in PROVIA improves coordination between stakeholders. Role-based workflows help ensure that proposals and evaluations pass through proper approval channels.

The structured workflow improves:

- Communication
- Monitoring
- Accountability
- Administrative efficiency

5.2 PHP and MySQL vs. Client Side Storage

PHP and MySQL were selected because they provide secure server-side processing and centralized data management.

Compared to client-side storage:

PHP & MySQL	Client-Side Storage
Centralized management	Limited scalability
Better security	Lower security
Multi-user support	Single-device dependency
Reliable database operations	Limited data control

The server-side architecture makes PROVIA more scalable and secure.

5.3 Limitations and Known Issues

Although the system performs efficiently, some limitations still exist:

- Limited real-time notifications
- Basic chat functionality
- Dependence on internet connectivity
- Limited mobile optimization in some modules

These issues can be improved in future versions.

5.4 Comparison with Existing Solutions

Compared to manual systems and spreadsheet-based management, PROVIA provides:

- Better organization
- Faster processing
- Centralized storage
- Improved transparency
- Secure access control

The system significantly reduces administrative workload and improves overall workflow management.

6. Conclusions and Recommendations

6.1 Conclusions

PROVIA successfully achieved its objectives by providing a centralized and secure PBL project management platform.

The system improved:

- Workflow efficiency
- Communication between stakeholders
- Project tracking
- Record management
- Proposal handling

The use of PHP and MySQL enabled the development of a scalable and maintainable web application.

Overall, the project demonstrates how web technologies can modernize academic project management processes.

6.2 Recommendations for Future Development

Future improvements may include:

- Real-time notifications
- Mobile application integration
- AI-based project recommendations
- Advanced analytics dashboard
- Video meeting integration
- Improved chat system
- Cloud deployment
- Automated report generation

These features would further improve usability and system performance.

7. References

1. PHP Official Documentation.
2. MySQL Documentation.
3. Bootstrap Framework Documentation.
4. JavaScript MDN Documentation.
5. Software Engineering and SDLC references.

6. Web Development Tutorials and Resources.

Appendices

Appendix A – Technologies Used

- PHP
- MySQL
- HTML5
- CSS3
- Bootstrap
- JavaScript
- Apache Server
- XAMPP

Appendix B – System Modules

- Authentication Module
- Proposal Management Module
- Evaluation Module
- Dashboard Module
- Chat and Communication Module
- User Management Module
- Reporting Module

Appendix C – System Features

- Role-based login system
- Proposal submission
- Supervisor assignment
- Evaluation tracking
- File uploads
- Dashboard analytics
- Feedback system
- Notification management